Absolute error

1 State the absolute error of the following:

- **a** A measurement of 30 kg, which was measured to the nearest kilogram.
- **b** A measurement of 400 g, which was measured to the nearest gram.
- **c** The amount $\$700$, which was calculated to the nearest $\$10$.

2 A chef measures the amount of olive oil to be used in a cup marked in millilitres. Find the absolute error of any measurement she makes.

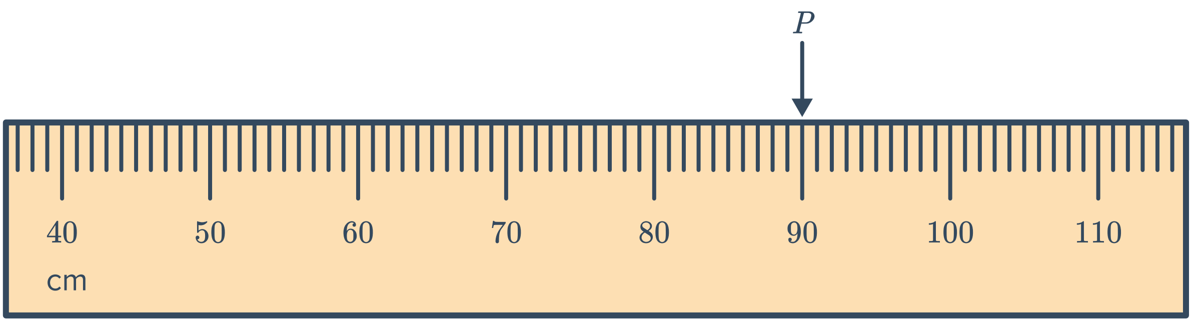
3 State the absolute error of each of the following measurements obtained from measuring devices:

- **a** 18.9 mL
- **b** 269 mm
- **c** 9.9 g
- **d** 18 s
- **e** 15.12 kg
- **f** 17.8 m
- **g** $117.38 ^\circ\text{C}$

4 The number of people at a concert is approximately 5050. Explain why we cannot find the absolute error of this approximation.

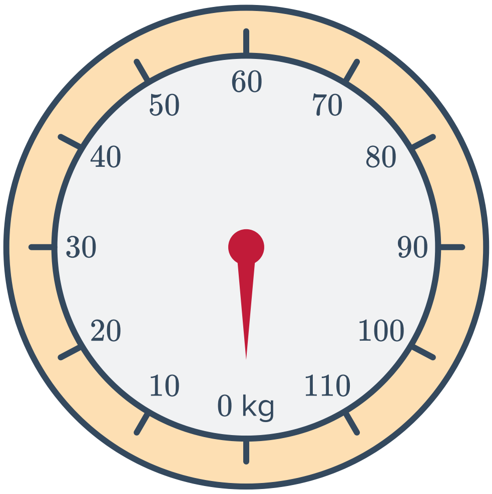
The limits of accuracy

5 Consider the tape measure below:



- **a** What is the smallest unit marked on the tape?
- **b** What should be recorded for measurement P?
- **c** What is the absolute error of measurement P?
- **d** State whether the following could be the actual measurement of P:
  - **i** 89.02 cm
  - **ii** 90.64 cm
  - **iii** 89 cm
  - **iv** 90.03 cm

6 Consider the following scale:



a What is the smallest unit labelled on the scale?

b What is the absolute error?

c If a certain object is measured at $70 \mathrm{~kg}$, what are the lower and upper bounds of this measurement?

7 Consider the speedometer below:

a At what speed is being indicated on the speedometer?

b What is the absolute error of this speed reading?

c What would the maximum speed limit need to be to ensure the driver is not speeding?

8 For the following measurements, find:

i The upper bound

ii The lower bound

a A distance measured to be $13.45 \mathrm{~km}$.

b A distance measured to be $6.4 \mathrm{~km}$.

c A height measured to be $5 \mathrm{~m}$.

d A bag of sugar weighs $14 \mathrm{~kg}$ to the nearest 10 grams.

e Puncak Jaya, Indonesia’s highest mountain, is $4884 \mathrm{~m}$ high rounded to the nearest metre.

f The cost of a CD lie if it is known to be $\$50$ correct to the nearest $\$5$.

9 The length of a piece of rope is measured to be $19.99 \mathrm{~m}$ using a ruler. What is the upper bound of the largest possible length of this rope?

10 The height of a tower is measured as 2.1 [redacted]at is the shortest possible height of the tower?

11 State the maximum and minimum possible number of the following:

- **a** A town's population is estimated to be 170 people, to the nearest 10 people.
- **b** A stack of paper is estimated to have 3000 sheets to the nearest 100 sheets.

Further calculations (Extended)

12 $c = 4$ and $d = 8$ have both been rounded to the nearest whole number.

- **a** Calculate the lower bound of the value of $c \times d$.
- **b** Calculate the upper bound of the value of $\frac{c}{d}$.

13 A school field has a 100 m track that is measured to the nearest metre. The record for the fastest 100 m sprint is 13.9 seconds rounded to the nearest 0.1 seconds.

- **a** What is the fastest speed, in metres per second, the record holder could have run? Round your answer correct to two decimal places.
- **b** What is the slowest speed, in metres per second, the record holder could have run? Round your answer correct to two decimal places.

14 A circle has a radius of 7 cm, correct to the nearest centimetre.

- **a** Calculate its area correct to one decimal place.
- **b** What are the smallest and largest possible areas? Round your answer to one decimal place.
- **c** What is the maximum possible absolute error in the calculation?
- **d** What is the maximum possible percentage error in the calculation? Round your answer to one decimal place.

15 A field has dimensions $15.4\text{ m} \times 17.6\text{ m}$, to the nearest 10 cm.

- **a** What are the upper and lower bounds of the area of the field?
- **b** What is the upper and lower bounds of the perimeter of the field?

16 A cube has a side length of $10\ \mathrm{cm}$ rounded to the nearest centimetre.

- **a** Find the lower and upper bounds of the side length.
- **b** Find the area of one face of the cube, using the given side length.
- **c** Find the lower and upper bounds of the area of one face.
- **d** Find the volume of the cube using the given side length.
- **e** Find the lower and upper bounds of the volume of the cube.